

Tak Yeon Lee

ASSISTANT PROFESSOR · DEPARTMENT OF INDUSTRIAL DESIGN

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Education

University of Maryland

College Park, MD, USA

PH.D COMPUTER SCIENCE

2009.1 - 2017.5

- Thesis: Toward Symbiotic Human-AI Interaction Focusing on Programming by Example
- Advisor: Prof. Benjamin B. Bederson

Delft University of Technology

Delft, Netherlands

MSc. INDUSTRIAL DESIGN ENGINEERING

2004.9 - 2006.11

- Advisor: Prof. Pieter Jan Stappers

Korea Advanced Institute of Science and Technology

Daejeon, Republic of Korea

BSc. INDUSTRIAL DESIGN

1996.3 - 2004.2

- Minors in Computer Science

Professional Experience

- 2021.3 – Current **Assistant Professor**, Department of Industrial Design, KAIST, Republic of Korea
- 2017.6 – 2021.2 **Research Scientist**, Adobe Research, San Jose, USA
- 2012.6 – 2012.8 **Research Intern**, IBM Research, Cambridge, MA, USA
- 2010.6 – 2010.8 **UX Design Intern**, Microsoft, Redmond, WA, USA
- 2009.3 – 2017.6 **Graduate Research Assistant**, University of Maryland, MD, USA

Publications

JOURNALS

- 1 J. Park, E. Park, D. Lee, S. Kang, **Takyeon Lee**, H. Hong, S.-J. Lee, H. Kim, Y. R. Park, and U. Lee, “Human digital twins for pervasive healthcare: A scoping review,” *Health Informatics Journal*, vol. 31, no. 4, p. 14 604 582 251 392 455, 2025, PMID: 41166748. DOI: 10 . 1177 / 14604582251392455.
- *2 Y. S. Kim, H. S. Moon, S. Lee, and **Lee, Tak Yeon**, “‘histochat’: Leveraging ai-driven historical personas for personalized and engaging middle school history education,” *Proc. ACM Hum.-Comput. Interact.*, vol. 9, no. 7, Oct. 2025. DOI: 10 . 1145 / 3757534.
- *3 **Tak Yeon Lee**, A. Smith, K. Seppi, N. Elmqvist, J. Boyd-Graber, and L. Findlater, “The human touch: How non-expert users perceive, interpret, and fix topic models,” *International Journal of Human-Computer Studies*, vol. 105, pp. 28–42, 2017, ISSN: 1071-5819. DOI: <https://doi.org/10.1016/j.ijhcs.2017.03.007>.

CONFERENCE PROCEEDINGS

- *1 Y. Jung, T. Hoang Anh Vo, H. S. Moon, J. Y. Choi, H. Oh, U. Lee, E. Kim, **Lee, Tak Yeon†**, and U. Lee, “Toward Flexible Psychiatric History-Taking and Visualization: Exploring Clinician Perspectives with Large Language Models,” en, in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona Spain: ACM, Apr. 2026, pp. 1–23, ISBN: 979-8-4007-2278-3. DOI: 10 . 1145 / 3772318 . 3790970.

*: Published at venues that are qualified as S-class or Top Conferences at KAIST

†: Co-corresponding author

- *2 S. Park, **Lee, Tak Yeon**[†], and W. Lee, “LLM-box vs. Thinking-box: Designing for Deliberate User Engagement with Distorted Information in Conversational Search,” en, in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona Spain: ACM, Apr. 2026, pp. 1–20, ISBN: 979-8-4007-2278-3. doi: 10.1145/3772318.3790271.
- 3 H. Lee, Y. Oh, H. S. Moon, and **Lee, Tak Yeon**, “Understanding How Creators Collaborate with Conversational Agents in Short-Form Video Production,” en, in *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona, Spain: ACM, Apr. 2026, pp. 1–6, ISBN: 979-8-4007-2281-3. doi: 10.1145/3772363.3798927.
- *4 S. G. Kim, J. Y. Choi, P. Y. Lee, J. Chung, R. Rossi, J. Kil, E. Koh, and **Lee, Tak Yeon**, “Grounded generation of embellished bar chart ensuring chart integrity,” in *2025 IEEE Visualization and Visual Analytics (VIS)*, 2025, pp. 101–105. doi: 10.1109/VIS60296.2025.00026.
- 5 J. Y. Choi, S. G. Kim, J. Jeong, R. A. Rossi, J. Kil, and **Lee, Tak Yeon**, “Gaze2prompt: Turning eye-tracking data into visual prompts for multimodal llms,” in *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, ser. UbiComp Companion ’25, Finland: Association for Computing Machinery, 2026, pp. 110–114, ISBN: 9798400714771. doi: 10.1145/3714394.3754401.
- *6 R. Luera, R. Rossi, F. DERNONCOURT, A. Siu, S. Kim, T. Yu, R. Zhang, X. Chen, N. Lipka, Z. Zhang, S. Gyeom Kim, and T. Yeon Lee, “Personalizing data delivery: Investigating user characteristics and enhancing llm predictions,” in *Companion Proceedings of the ACM on Web Conference 2025*, ser. WWW ’25, Sydney NSW, Australia: Association for Computing Machinery, 2025, pp. 1167–1171, ISBN: 9798400713316. doi: 10.1145/3701716.3715452.
- 7 J. Chung, S. G. Kim, and **Lee, Tak Yeon**, “Wrightthere: Supporting children’s creative writing with ai-infused interactive 3d environment,” in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA ’25, Association for Computing Machinery, 2025, ISBN: 9798400713958. doi: 10.1145/3706599.3719932.
- 8 J. Jeong and **Lee, Tak Yeon**, “Ligs: Developing an llm-infused game system for emergent narrative,” in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA ’25, Association for Computing Machinery, 2025, ISBN: 9798400713958. doi: 10.1145/3706599.3720212.
- 9 H. Bae, N. Kim, S. Lee, and **Lee, Tak Yeon**, “Bridging bond beyond life: Designing vr memorial space with stakeholder collaboration via research through design,” in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA ’25, Association for Computing Machinery, 2025, ISBN: 9798400713958. doi: 10.1145/3706599.3719871.
- 10 S. G. Kim, J. Y. Choi, R. Rossi, E. Koh, and **Lee, Tak Yeon**, “Chart-to-experience: Benchmarking multimodal llms for predicting experiential impact of charts,” in *2025 IEEE 18th Pacific Visualization Conference (PacificVis)*, 2025, pp. 340–345. doi: 10.1109/PacificVis64226.2025.00040.
- 11 H. Im, T. Kim, E. Jung, B. Ku, S. Baek, Y.-k. Lim, T.-J. Nam, and **Lee, Tak Yeon**, “Virfie: Enhancing remote togetherness with user-created scenarios for virtual group selfie,” in *Intelligent Systems and Applications*, K. Arai, Ed., Cham: Springer Nature Switzerland, 2024, pp. 628–645, ISBN: 978-3-031-66336-9.
- *12 J. Han, H. Yoo, J. Myung, M. Kim, **Lee, Tak Yeon**, S.-Y. Ahn, and A. Oh, “RECIPE4U: Student-ChatGPT interaction dataset in EFL writing education,” in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, Eds., Torino, Italia: ELRA and ICCL, May 2024, pp. 13 666–13 676.
- 13 C. Harris, R. A. Rossi, S. Malik, J. Hoffswell, F. Du, **Lee, Tak Yeon**, E. Koh, and H. Zhao, “Visual insight recommendation: From ranking insight visualizations to insight types,” in *2023 IEEE International Conference on Big Data (BigData)*, 2023, pp. 1716–1725. doi: 10.1109/BigData59044.2023.10386099.
- 14 J. Han, H. Yoo, Y. Kim, J. Myung, M. Kim, H. Lim, J. Kim, **Lee, Tak Yeon**, H. Hong, S.-Y. Ahn, and A. Oh, “Recipe: How to integrate chatgpt into efl writing education,” in *Proceedings of the Tenth ACM Conference on Learning Scale*, ser. LS ’23, Copenhagen, Denmark: Association for Computing Machinery, 2023, pp. 416–420, ISBN: 9798400700255. doi: 10.1145/3573051.3596200.
- *15 C. Harris, R. Rossi, S. Malik, J. Hoffswell, F. Du, **Lee, Tak Yeon**, E. Koh, and H. Zhao, “Spotlight: Visual insight recommendation,” in *Companion Proceedings of the ACM Web Conference 2023*, ser. WWW ’23 Com-

- panion, Austin, TX, USA: Association for Computing Machinery, 2023, pp. 19–23, ISBN: 9781450394192. DOI: 10.1145/3543873.3587302.
- *16 X. Qian, R. A. Rossi, F. Du, S. Kim, E. Koh, S. Malik, **Lee, Tak Yeon**, and N. K. Ahmed, “Personalized visualization recommendation,” 3, vol. 16, New York, NY, USA: Association for Computing Machinery, Sep. 2022. DOI: 10.1145/3538703.
 - 17 H. Im, T. Kim, E. Jung, B. Ku, S. Baek, and **Lee, Tak Yeon**, “Virfie: Virtual group selfie station for remote togetherness,” in *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems*, ser. CHI EA ’22, New Orleans, LA, USA: Association for Computing Machinery, 2022, ISBN: 9781450391566. DOI: 10.1145/3491101.3519767.
 - *18 Z. Zeng, P. Moh, F. Du, J. Hoffswell, **Lee, Tak Yeon**, S. Malik, E. Koh, and L. Battle, “An evaluation-focused framework for visualization recommendation algorithms,” 1, vol. 28, 2022, pp. 346–356. DOI: 10.1109/TVCG.2021.3114814.
 - *19 X. Qian, R. A. Rossi, F. Du, S. Kim, E. Koh, S. Malik, **Lee, Tak Yeon**, and J. Chan, “Learning to recommend visualizations from data,” in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, ser. KDD ’21, Virtual Event, Singapore: Association for Computing Machinery, 2021, pp. 1359–1369, ISBN: 9781450383325. DOI: 10.1145/3447548.3467224.
 - *20 Y. Xian, H. Zhao, **Lee, Tak Yeon**, S. Kim, R. Rossi, Z. Fu, G. de Melo, and S. Muthukrishnan, “Exacta: Explainable column annotation,” in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, ser. KDD ’21, Virtual Event, Singapore: Association for Computing Machinery, 2021, pp. 3775–3785, ISBN: 9781450383325. DOI: 10.1145/3447548.3467211.
 - *21 X. Qian, E. Koh, F. Du, S. Kim, J. Chan, R. A. Rossi, S. Malik, and **Lee, Tak Yeon**, “Generating accurate caption units for figure captioning,” in *Proceedings of the Web Conference 2021*, ser. WWW ’21, Ljubljana, Slovenia: Association for Computing Machinery, 2021, pp. 2792–2804, ISBN: 9781450383127. DOI: 10.1145/3442381.3449923.
 - 22 N. Lipka, **Lee, Tak Yeon**, and E. Koh, “Detecting link and landing page misalignment in marketing emails,” in *Database and Expert Systems Applications*, M. Elloumi, M. Granitzer, A. Hameurlain, C. Seifert, B. Stein, A. M. Tjoa, and R. Wagner, Eds., Cham: Springer International Publishing, 2018, pp. 254–264, ISBN: 978-3-319-99133-7.
 - 23 **Lee, Tak Yeon** and E. Koh, “Identifying types of misalignments between promotion emails and landing pages,” in *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, ser. CHI EA ’18, Montreal QC, Canada: Association for Computing Machinery, 2018, pp. 1–6, ISBN: 9781450356213. DOI: 10.1145/3170427.3188684.
 - 24 **Lee, Tak Yeon**, C. Dugan, and B. B. Bederson, “Towards understanding human mistakes of programming by example: An online user study,” in *Proceedings of the 22nd International Conference on Intelligent User Interfaces*, ser. IUI ’17, Limassol, Cyprus: Association for Computing Machinery, 2017, pp. 257–261, ISBN: 9781450343480. DOI: 10.1145/3025171.3025203.
 - *25 A. Smith, **Lee, Tak Yeon**, F. Poursabzi-Sangdeh, J. Boyd-Graber, N. Elmqvist, and L. Findlater, “Evaluating visual representations for topic understanding and their effects on manually generated topic labels,” L. Lee, M. Johnson, and K. Toutanova, Eds., vol. 5, Cambridge, MA: MIT Press, 2017, pp. 1–16. DOI: 10.1162/tac1_a_00042.
 - 26 **Lee, Tak Yeon** and B. B. Bederson, “Give the people what they want: Studying end-user needs for enhancing the web,” vol. 2, Nov. 2016, e91. DOI: 10.7717/peerj-cs.91.
 - 27 **Tak Yeon Lee**, M. L. Mauriello, J. Ahn, and B. B. Bederson, “Ctarcade: Computational thinking with games in school age children,” 1, vol. 2, 2014, pp. 26–33. DOI: <https://doi.org/10.1016/j.ijcci.2014.06.003>.
 - 28 **Lee, Tak Yeon**, C. Dugan, W. Geyer, T. Ratchford, J. Rasmussen, N. S. Shami, and S. Lupushor, “Experiments on motivational feedback for crowdsourced workers,” 1, vol. 7, Aug. 2013, pp. 341–350. DOI: 10.1609/icwsm.v7i1.14428.
 - 29 **Lee, Tak Yeon**, M. L. Mauriello, J. Ingraham, A. Sopan, J. Ahn, and B. B. Bederson, “Ctarcade: Learning computational thinking while training virtual characters through game play,” in *CHI ’12 Extended Abstracts*

on *Human Factors in Computing Systems*, ser. CHI EA '12, Austin, Texas, USA: Association for Computing Machinery, 2012, pp. 2309–2314, ISBN: 9781450310161. DOI: 10.1145/2212776.2223794.

- 30 M. Rios-Berrios, P. Sharma, **Tak Yeon Lee**, R. Schwartz, and B. Shneiderman, “Treecover: Coordinated dual treemap visualization for exploring the recovery act,” 2, vol. 29, 2012, pp. 212–222. DOI: <https://doi.org/10.1016/j.giq.2011.07.004>.

Awards

- 2026 **WrightHere**, iF Design Award
2025 **Chart-to-Experience**, Best VisNote, IEEE PacificVis
2024 **Dear.**, iF Design Award

Patents

- Jan 2026 **Machine learning techniques for generating visualization recommendations**, Patent, US 19343670
Jun 2024 **Systems for generating interactive reports**, Patent, US 17474188
May 2024 **Lifelong schema matching**, Patent, US 17036453
Dec 2023 **Facilitating generation of data visualizations via natural language processing**, Patent, US 17354954
Oct 2023 **Configuration of user interface for intuitive selection of insight visualizations**, Patent, US 17161770
Oct 2023 **Generating visual data stories**, Patent, US 18069561
Oct 2023 **Automated caption generation from a dataset**, Patent, US 17094435
Aug 2023 **Personalized visualization recommendation system**, Patent, US 17091941
May 2023 **Generating explanatory paths for predicted column annotations**, Patent, US 16796681
Feb 2023 **Dynamic user interface and machine learning tools for generating digital content and multivariate testing recommendations**, Patent, US 17383114
Jan 2023 **Dynamically determining schema labels using a hybrid neural network encoder**, Patent, US 16751755
Oct 2022 **Detecting and resolving semantic misalignments between digital messages and external digital content**, Patent, US 15910926
May 2022 **Identifying and presenting misalignments between digital messages and external digital content**, Patent, US 16419676
Mar 2022 **Graph-based configuration of user interface for selection of features in visualization applications**, Patent, US 17015495
Oct 2021 **Systems for suggesting content components**, Patent, US 17183055

Invited Talks

2023.03	How to Live in the Era of Generative AI, KAIST
2024.11	The Role of Designers in the Age of AI, KAIST AI Executive Program
2024.12	UX Designers in the Age of AI: Roles and Opportunities, LG
2024.11	Object-Oriented Design in the Age of AI, Design 3.0 Forum
2024.11	Large-scale Persona Generation and UX Testing with AI and Small Data, Korea University
2024.07	AI as a Design Material, LG
2024.03	AI Utilization Strategies for UX Designers, Samsung SDS
2024.02	AI Utilization Strategies for UX Designers, INITION
2023.08	AI and Visual Arts, Ministry of Culture, Sports and Tourism
2023.07	Understanding AI, Understanding Humans, Moontak Network
2023.04	How to Live in the Era of Generative AI, Korean Intellectual Property Office
2022.10	On Convergence: Global Conference, Ringle
2022.04	Designers in the Age of AI and Data, Korean Society of Design Convergence
2021.11	Design and Technology, Korea Institute of Design Promotion
2021.03	At the Intersection of Design and AI, Handong Global University
2021.07	Designing AI-infused Products for Smart Life, Seoul National University Medical School
2021.03	What if We Apply Data-Driven Decision Making to Product Design?, LG Electronics
2021.05	Toward Human-Centered AI for Healthcare Workers, Akasa (USA)

Teaching Experience

Spring 2026	Design Project 1 (ID.50003)	KAIST
Fall 2025	Data Analytics for Designers (ID.40018)	KAIST
Spring 2025	System Design (ID.40003)	KAIST
Spring 2025	Design Special Topics V: Designing Artificial Human Behavior and Experience (ID.49902)	KAIST
Fall 2024	Data Analytics for Designers (ID.40018)	KAIST
Spring 2024	System Design (ID.40003)	KAIST
Fall 2023	Design Special Topics V: Data Analytics for Designers (ID.49902)	KAIST
Spring 2023	System Design (ID.40003)	KAIST
Fall 2022	Design Special Topics V: Data Analytics for Designers (ID.49902)	KAIST
Fall 2022	Freshman Seminar: School of Industrial Design (HSS.10090)	KAIST
Spring 2022	System Design (ID.40003)	KAIST
Fall 2021	Smart Life: AI-based Service Design (CoE.49001)	KAIST
Fall 2021	Design Special Topics V: Data Analytics for Designers (ID.49902)	KAIST
Spring 2021	Design Project (ID.50003)	KAIST

Service and Outreach

2026 – 2027	PacificVis 2027 , Conference Sponsor Chair	<i>Busan, Korea</i>
2023	“AI Design Park” Strategic Planning Initiative , Lead Advisory Committee Member	<i>KEIT</i>
2024	Design Innovation Technology Roadmap Initiative , Advisory Committee Member	<i>MOTIE</i>
2022	SIGCHI , Local Event Moderator	<i>Seoul, Korea</i>
2021 – 2022	TEI 2022 , Conference Local Chair	<i>Daejeon, Korea</i>